

Git – Merge Conflicts

So far we've had no merge conflicts

- When might we have merge conflict?
 - People editing the same line
 - file both edited and deleted

```
$ git merge dev
Auto-merging main.txt
CONFLICT (content): Merge conflict in main.txt
Automatic merge failed; fix conflicts and then commit the result.
```

Resolving merge commits (1)

- We have to manually edit files with conflicts
 - we decide what to keep
 - each conflicted marked by markers:

```
# No conflict
<<<<<<<
This is from local branch (one we're merging into)
=====
This is from remote (one being merged in)
>>>>>>>
# Conflict over
```

Resolving merge commits (1.5)

- git status will show which files have conflicts

```
On branch merge_test_1
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add/rm <file>..." as appropriate to mark resolution)
    both modified:   stefan.txt
    deleted by them: test.txt
```

Resolving merge commits (2)

- Decide what to keep
 - Can be one version or a mixture
- Edit the file to match
- Remove markers
- Repeat for all conflicts
- Stage and commit files

Resolving merge commits (3)

- shortcut for keeping one version
 - `git restore --ours file`
 - `git restore --theirs file`
 - ours = branch being merged into, theirs is opposite
- what if we mess up?
 - `git merge --abort`
- can delete changes in working directory